LESSON 01 / 08

Why one number controls the whole subject

A **quadratic equation** is $ax^2 + bx + c = 0$ with $a \neq 0$ (if $a = 0$ it is linear, and every formula below collapses). Coefficients are real unless stated otherwise. The equation asks: where does the parabola $y = ax^2 + bx + c$ meet the x-axis? Everything — nature of roots, sign of the expression, location of roots — is bookkeeping built on one quantity.

SHORT DERIVATION

Completing the square forces the formula and the discriminant

Divide by a and complete the square:

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \quad \Rightarrow \quad \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

The left side is a square, so a real solution exists iff the right side is ≥ 0, i.e. iff $D = b^2 - 4ac \geq 0$. Taking roots gives the **quadratic formula**

$$x = \frac{-b \pm \sqrt{D}}{2a}, \qquad D = b^2 - 4ac$$

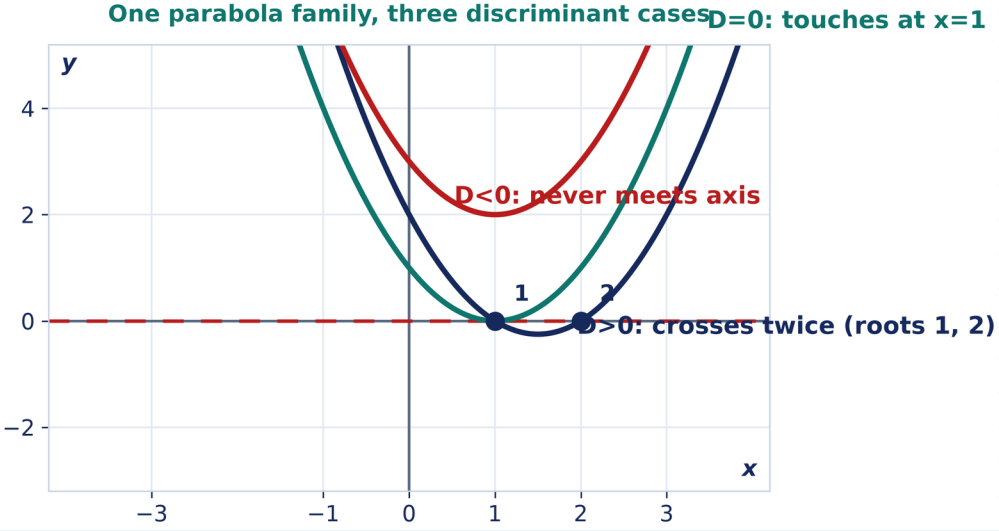
D is called the **discriminant** because it discriminates between the cases. Note it is $b^2 - 4ac$, never $b^2 + 4ac$.

KEY RESULT

The three cases — memorise with the picture

- $D > 0$: two **distinct real** roots; the parabola crosses the axis twice.
- $D = 0$: one **repeated** real root $x = -\frac{b}{2a}$; the parabola touches the axis at its vertex.
- $D < 0$: **no real roots**, but two complex roots, always a conjugate pair $\alpha \pm i\beta$ (real coefficients force this: conjugating the equation changes nothing, so $\bar{\alpha}$ is a root whenever α is).

Saying "$D < 0$ means no roots" is wrong — it means no real roots.



WORKED EXAMPLE

Worked — reading the case off D

$x^2 - 5x + 6 = 0$: $D = 25 - 24 = 1 > 0$, so $x = \frac{5 \pm 1}{2}$, giving $x = 2, 3$.

$2x^2 - 3x + 5 = 0$: $D = 9 - 40 = -31 < 0$, so $x = \frac{3 \pm i\sqrt{31}}{4}$ — non-real conjugates.

Equal roots: $x^2 + kx + 9 = 0$ needs $D = k^2 - 36 = 0$, so $k = \pm 6$ (both signs — a classic one-mark loss).

Rational, irrational, and conjugate root pairs

KEY RESULT

What the coefficients force the roots to look like

- Rational coefficients and D a perfect square (of a rational) $\Leftrightarrow$ both roots **rational**.
- $D > 0$ but not a perfect square: roots are irrational **conjugates** $p \pm \sqrt{q}$ with p, q rational.
- Real coefficients, $D < 0$: complex conjugates $\alpha \pm i\beta$.
- One root known of conjugate type forces the other: if $2 + \sqrt{3}$ is a root of a rational-coefficient quadratic, $2 - \sqrt{3}$ is the other root. Product $= 4 - 3 = 1$, sum $= 4$, so the equation is $x^2 - 4x + 1 = 0$.

COMMON TRAP

Common traps — foundations

- $D = b^2 - 4ac$: the minus sign, and the 4.
- "Equal roots" means $D = 0$ **and** the equation is genuinely quadratic ($a \neq 0$). If a contains the parameter, check the degenerate case separately.
- If coefficients depend on a parameter, $D \geq 0$ alone does not guarantee two real roots when the equation can degenerate to linear.
- A repeated root is still a root of multiplicity two — it counts twice in Vieta's relations.

ASSUMPTIONS & CONVENTIONS

Standing conventions for this chapter

$a, b, c \in \mathbb{R}$, $a \neq 0$; roots denoted α, β; $S = \alpha + \beta$, $P = \alpha\beta$; "roots" means complex roots counted with multiplicity unless "real roots" is said. All root-location conditions in sheet 04 assume real coefficients.

LESSON 02 / 08

The two relations that do all the work

SHORT DERIVATION

Vieta from factorising

If α, β are the roots, then $ax^2+bx+c = a(x-\alpha)(x-\beta) = a(x^2-(\alpha+\beta)x+\alpha\beta)$. Comparing coefficients:

$$S = \alpha+\beta = -\frac{b}{a}, \qquad P = \alpha\beta = \frac{c}{a}$$

Watch the signs: sum carries the minus, product does not. The derivation also shows why Vieta survives in $\mathbb{C}$: factorisation needs no reality.

KEY RESULT

The symmetric-function toolkit (all from S and P)

$$\alpha^2+\beta^2 = S^2-2P, \qquad \alpha^3+\beta^3 = S^3-3PS$$

$$\alpha^2\beta+\alpha\beta^2 = PS, \qquad \frac{1}{\alpha}+\frac{1}{\beta} = \frac{S}{P} \quad (P\neq 0)$$

$$(\alpha-\beta)^2 = S^2-4P = \frac{D}{a^2}, \qquad |\alpha-\beta| = \frac{\sqrt{D}}{|a|} \ (D\geq 0)$$

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha} = \frac{S^2-2P}{P}, \qquad \alpha^2-\beta^2 = (\alpha-\beta)S$$

Deep reason: swapping $\alpha \leftrightarrow \beta$ changes no coefficient, so any symmetric expression must be expressible through S, P alone. $\alpha-\beta$ is **not** symmetric — that is exactly why the discriminant has to be brought in for it.

WORKED EXAMPLE

Worked — everything from S and P, roots never found

For $x^2-5x+6=0$: $S=5$, $P=6$. Then $\alpha^2+\beta^2 = 25-12 = 13$; $\alpha^3+\beta^3 = 125-90 = 35$; $\frac{1}{\alpha}+\frac{1}{\beta} = \frac{5}{6}$; $(\alpha-\beta)^2 = 25-24 = 1$; $\alpha^2\beta+\alpha\beta^2 = 30$; $\frac{\alpha}{\beta}+\frac{\beta}{\alpha} = \frac{13}{6}$. (Roots are $2, 3$ — check: $4+9=13$, $8+27=35$ ✓.)

For $3x^2-7x+2=0$: $S=\frac{7}{3}$, $P=\frac{2}{3}$ directly — solving $(x=2, \frac{1}{3})$ is wasted effort.

COMMON TRAP

Common traps — Vieta

- $(\alpha-\beta)^2 = S^2-4P$, not S^2-2P (that is $\alpha^2+\beta^2$). The cross-term doubles.
- $|\alpha-\beta| = \sqrt{S^2-4P}$ needs $D\geq 0$ to be a real distance; over $\mathbb{C}$ only $(\alpha-\beta)^2$ is meaningful.
- Reciprocal identities need $P\neq 0$, i.e. $c\neq 0$: a zero root has no reciprocal.
- $\alpha-\beta$ has a sign ambiguity; problems asking for it usually want $|\alpha-\beta|$ or $(\alpha-\beta)^2$.

Forming an equation from its roots

KEY RESULT

The reverse direction

Roots $r_1, r_2 \Rightarrow$ equation $x^2 - (r_1 + r_2)x + r_1 r_2 = 0$ (monic form; multiply out to clear fractions). This is the whole of "form the equation whose roots are …": compute the **new sum** and **new product**, never the roots themselves.

Roots α^2, β^2 for $x^2 - 5x + 3 = 0$ $(S = 5, P = 3)$: new sum $= S^2 - 2P = 19$, new product $= P^2 = 9$, so $x^2 - 19x + 9 = 0$. Note product P^2, not P — checklist item.

STANDARD APPLICATION

Where this is used everywhere else

Condition problems ("roots in ratio $m : n$", "one root is the square of the other", "roots differ by k") are all Vieta plus one extra equation. E.g. roots differing by 1 for $x^2 - 5x + 6$: $(\alpha - \beta)^2 = S^2 - 4P = 1$ gives the difference without solving. Ratio $m : n$: set roots mt, nt, then $S = (m + n)t$ and $P = mn\,t^2$, so $P(m + n)^2 = mn\,S^2$ — eliminate t immediately.

LESSON 03 / 08

The expression as a function

SHORT DERIVATION

Completed-square form reads off everything

$$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2-\frac{D}{4a}$$

The square term is ≥ 0, so the extremum sits at the **vertex** $x_V=-\frac{b}{2a}$ and equals $-\frac{D}{4a}=\frac{4ac-b^2}{4a}$: a **minimum** if $a>0$, a **maximum** if $a<0$. The axis of symmetry is $x=-\frac{b}{2a}$.

WORKED EXAMPLE

Worked — vertex without calculus

$x^2-6x+11=(x-3)^2+2$: minimum 2 at $x=3$. $-x^2+4x-1=-(x-2)^2+3$: maximum 3 at $x=2$. For $y=2x^2-8x+5$: $x_V=\frac{8}{4}=2$, $y_V=\frac{40-64}{8}=-3$; vertex $(2,-3)$, axis $x=2$.

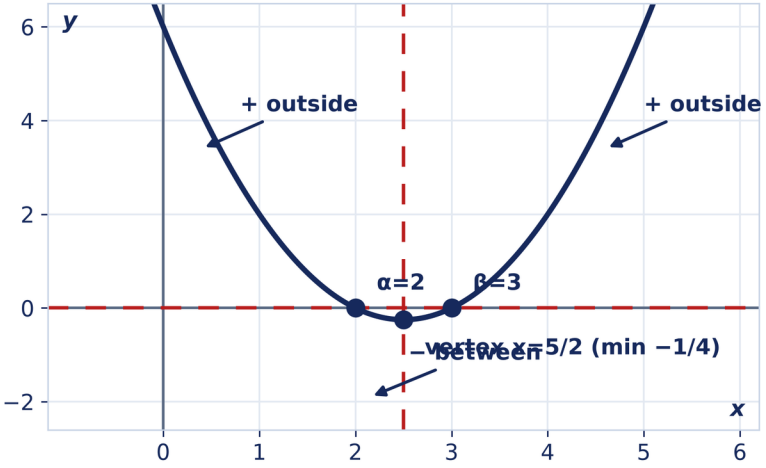
KEY RESULT

Sign analysis — the picture that solves every inequality

With real roots $\alpha<\beta$: for $a>0$ the expression is **positive outside** $[\alpha,\beta]$ and **negative between** them ("outside positive, inside negative" for an upward parabola). For $a<0$ all signs reverse. If $D<0$ the sign never changes: it is the sign of a everywhere. Solve $ax^2+bx+c>0$ by roots-then-intervals, never by testing a random middle point. $x^2-5x+6<0 \Leftrightarrow 2<x<3$.

y = x²−5x+6 (a>0)

Sign of a quadratic: + outside the roots, − between (a>0)



KEY RESULT

Never-negative / always-positive conditions

$ax^2 + bx + c \geq 0$ for **all** real $x \Leftrightarrow a > 0$ **and** $D \leq 0$. Similarly ≤ 0 for all $x \Leftrightarrow a < 0$ and $D \leq 0$. Both conditions are needed: $D \leq 0$ with $a < 0$ gives an always-negative expression. This pair is the standard engine for parameter ranges and later powers the discriminant method for ranges of rational functions.

Range on all of $\mathbb{R}$: $a > 0$: $\left[\frac{4ac - b^2}{4a}, \infty\right)$; $a < 0$: $\left(-\infty, \frac{4ac - b^2}{4a}\right]$. On a restricted interval, check whether the vertex lies inside; if not, the extremum is at an endpoint.

COMMON TRAP

Common traps — the expression

- Calling $\frac{4ac - b^2}{4a}$ a maximum when $a > 0$ — it is a minimum; the sign of a decides.
- Sign pattern quoted without the sign of a.
- "Always positive" written with $D < 0$ but strict/non-strict mixed up: > 0 for all x needs $a > 0, D < 0$; ≥ 0 allows $D = 0$ (touching zero at one point).
- On interval range problems, using the vertex value when the vertex is outside the interval.

INTUITION

AM-GM is this sheet in disguise

For $x > 0$, $x + \frac{k}{x} \geq 2\sqrt{k}$ is $(\sqrt{x} - \frac{\sqrt{k}}{\sqrt{x}})^2 \geq 0$ — a perfect square, i.e. a $D = 0$ quadratic. Least value of $x + \frac{4}{x}$ is 4 at $x = 2$. Most "quadratic-flavoured" extrema are a vertex or a square in hiding.

LESSON 04 / 08

The three ingredients

Let $f(x) = ax^2 + bx + c$, $a \neq 0$, roots $\alpha \leq \beta$ real. Every location condition is assembled from three ingredients only:

KEY RESULT

The location toolkit

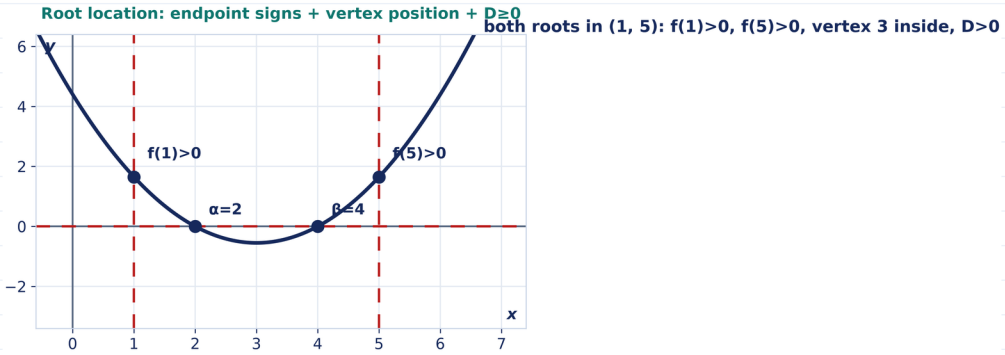
1. **Reality:** $D \geq 0$.
2. **Vertex position:** $x_v = -\frac{b}{2a}$ relative to the interval (the axis must sit where the roots are claimed to be).
3. **Endpoint sign:** $a \cdot f(k)$ at endpoints. The product $af(k) > 0$ means k lies *outside* $[\alpha, \beta]$ (same sign region as far away); $af(k) < 0$ means k lies *strictly between* the roots; $f(k) = 0$ means k **is** a root.

Multiply by a to kill case-splitting on the sign of a; always work with $af(k)$.

KEY RESULT

The standard configurations (learn as pictures, not formulas)

- **Both roots** $> k$: $D \geq 0$, $af(k) > 0$, $x_v > k$.
- **Both roots** $< k$: $D \geq 0$, $af(k) > 0$, $x_v < k$.
- k **between the roots** ($\alpha < k < \beta$): $af(k) < 0$ alone (this forces $D > 0$ automatically).
- **Both roots in** (k_1, k_2): $D \geq 0$, $af(k_1) > 0$, $af(k_2) > 0$, $k_1 < x_v < k_2$.
- **Exactly one root in** (k_1, k_2): $f(k_1)f(k_2) < 0$ (then $D > 0$ is automatic), plus separately check endpoint roots $f(k_i) = 0$.
- **Roots of opposite signs:** $af(0) = ac < 0$ alone (i.e. $P = \frac{c}{a} < 0$); again $D > 0$ is automatic.
- **Both roots positive:** $D \geq 0$, $S > 0$, $P > 0$. Both negative: $D \geq 0$, $S < 0$, $P > 0$. (This is the $k = 0$ case of the first two bullets — Vieta form.)



WORKED EXAMPLE

Worked — both roots exceed 1

Both roots of $x^2 - 2kx + k^2 - 1 = 0$ exceed 1. $D = 4 > 0$ always; the roots are $k \pm 1$, so the smaller exceeds 1 iff $k - 1 > 1$, i.e. $k > 2$. Toolkit cross-check: $x_v = k > 1$ and $af(1) = k^2 - 2k = k(k-2) > 0$ give $k > 2$ or $k < 0$; intersecting with $k > 1$ leaves $k > 2$ ✓. Solving directly was cleaner here only because the roots are explicit — use the toolkit when they are not.

WORKED EXAMPLE

Worked — exactly one root in (0, 2)

$f(x) = x^2 - 3x + k$ has exactly one root in $(0, 2)$: $f(0)f(2) = k(k-2) < 0$ gives $0 < k < 2$; endpoints must be checked individually: $k = 0$ gives roots $0, 3$ — none inside, excluded; $k = 2$ gives roots $1, 2$ — the root 1 **is** inside, so $k = 2$ is included. Answer $0 < k \leq 2$, with the endpoint discussion stated — losing it (in particular, mechanically excluding both endpoints) is the standard half-mark.

COMMON TRAP

Common traps — location

- Forgetting any one of the three ingredients: $D \geq 0$ and $af(k) > 0$ without the vertex condition also admits both roots on the other side of k.
- Using $f(k) > 0$ instead of $af(k) > 0$ when a may be negative or contains a parameter.
- Adding $D > 0$ redundantly (and then mishandling equality) when $af(k) < 0$ already forces distinct real roots.
- Strict vs non-strict: "in (k_1, k_2)" excludes endpoint roots; "in $[k_1, k_2]$" requires checking $f(k_i) = 0$ cases separately.
- When the leading coefficient contains the parameter, the degenerate linear case must be tested on its own.

ASSUMPTIONS & CONVENTIONS

Assumptions behind the toolkit

Real coefficients and real roots throughout; "root in interval" claims are meaningless for non-real roots. If a depends on the parameter, first confirm the equation stays quadratic on the claimed parameter range.

LESSON 05 / 08

The standard substitutions

KEY RESULT

Recognise the shape, name the substitution

- $ax^4 + bx^2 + c = 0$ (biquadratic): $t = x^2$, keep only $t \geq 0$.
- $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x + \frac{1}{x}\right) + c = 0$: $t = x + \frac{1}{x}$, using $x^2 + \frac{1}{x^2} = t^2 - 2$. Note $|t| \geq 2$ for real $x \neq 0$ — reject $|t| < 2$.
- $a^{2x} + b\,a^x + c = 0$ (and $\sin$, $\log$ analogues): $t = a^x > 0$ — require $t > 0$.
- $(x + a)(x + b)(x + c)(x + d) = k$ with $a + d = b + c$: pair outer with outer so both products share the same $x^2 + (\text{common})x$ part; set t to that shared part.
- Radical equations: isolate the radical, square, solve — then **verify every candidate in the original equation**.
- Anything of the form $g(x)^2 + b\,g(x) + c = 0$: $t = g(x)$, with the range of g as the admissibility filter.

WORKED EXAMPLE

Worked — the four canonical types

Biquadratic: $x^4 - 5x^2 + 4 = 0$: $t = x^2$ gives $t = 1, 4$, so $x = \pm 1, \pm 2$.

Exponential: $2^{2x} - 5 \cdot 2^x + 4 = 0$: $t = 2^x > 0$ gives $t = 1, 4$ (both admissible), so $x = 0, 2$.

Reciprocal: $x^2 + \frac{1}{x^2} = 7$: with $t = x + \frac{1}{x}$, $t^2 - 2 = 7$, $t = \pm 3$; then $x = \frac{3 \pm \sqrt{5}}{2}$ or $x = \frac{-3 \pm \sqrt{5}}{2}$.

Radical: $\sqrt{x + 3} = x - 3$: squaring gives $x^2 - 7x + 6 = 0$, candidates $x = 1, 6$. Check: $x = 1$ gives $2 = -2$ ✗ (also the RHS is negative while a square root never is — rejectable *before* squaring); $x = 6$ gives $3 = 3$ ✓. Answer: $x = 6$ only.

COMMON TRAP

Common traps — reducible equations

- Squaring is not reversible: extraneous roots are the rule, not the exception. Verification is part of the method.
- Dividing by an expression that can vanish loses roots: $(x + 1)^2 = 4(x + 1)$ divided by $x + 1$ loses $x = -1$; the substitution $t = x + 1$ gives $t = 0, 4$, i.e. $x = -1, 3$.
- Admissibility forgotten: $t = x^2 < 0$ or $t = 2^x \leq 0$ or $\left|x + \frac{1}{x}\right| < 2$ candidates must be discarded even though the quadratic "gave" them.
- For $\sqrt{A} = B$, record $B \geq 0$ and $A \geq 0$ first — often half the candidates die instantly.

STANDARD APPLICATION

Why quadratics are the reduction target

Biquadratics, radical, exponential, logarithmic and many trigonometric equations all collapse to a quadratic after one well-chosen substitution. The exam skill is not solving the quadratic — it is (1) spotting the repeated block $g(x)$, (2) writing its admissible range, (3) filtering. Marks are lost at steps 2–3, almost never at step 1.

LESSON 06 / 08

A common root

KEY RESULT

Subtract first, condition later

If $ax^2+bx+c=0$ and $a'x^2+b'x+c'=0$ share a root α, subtracting kills the x^2 term (after scaling to equalise leading coefficients): the common root satisfies a **linear** equation, and is usually immediate. The textbook condition

$$(bc'-b'c)^2=(ab'-a'b)(ac'-a'c)$$

is the same statement after elimination — correct but slow; reach for it only when subtraction degenerates (identical or proportional equations, which share *both* roots: coefficients proportional, $\frac{a}{a'}=\frac{b}{b'}=\frac{c}{c'}$).

WORKED EXAMPLE

Worked — common roots

$x^2-5x+6=0$ and $x^2-4x+3=0$: subtract, $-x+3=0$, so the common root is $x=3$ (check: $9-15+6=0$ ✓, $9-12+3=0$ ✓).

If $x^2+ax+b=0$ and $x^2+bx+a=0$ $(a\neq b)$ share a root: subtracting gives $(a-b)(x-1)=0$, so the common root is 1 and substituting gives $a+b+1=0$.

Transformed roots

KEY RESULT

Build from new S and new P — or substitute and skip the algebra

- Roots $\alpha+k,\beta+k$: new sum $S+2k$, new product $P+kS+k^2$. Equivalent shortcut: replace x by $x-k$ in the original equation.
- Roots $\frac{1}{\alpha},\frac{1}{\beta}$ $(P\neq 0)$: new sum $\frac{S}{P}$, new product $\frac{1}{P}$. Shortcut: reverse the coefficients — $cx^2+bx+a=0$.
- Roots $k\alpha,k\beta$: sum kS, product k^2P. Shortcut: replace x by $\frac{x}{k}$ and clear.
- Roots α^2,β^2: sum S^2-2P, product P^2.
- Roots α^3,β^3: sum S^3-3PS, product P^3.

WORKED EXAMPLE

Worked — transformations

From $x^2-5x+6=0$ $(S=5,P=6)$: roots $\alpha+1,\beta+1$ give sum 7, product $6+5+1=12$, equation $x^2-7x+12=0$ (actual roots $3,4$ ✓). Reciprocals give sum $\frac{5}{6}$, product $\frac{1}{6}$, equation $6x^2-5x+1=0$ (actual $\frac{1}{2},\frac{1}{3}$ ✓).

COMMON TRAP

Common traps — common and transformed roots

- The squared-roots equation has product P^2, not P.
- Reciprocal transformation silently assumes neither root is 0; if $c = 0$ the "reciprocal equation" drops degree.
- Common-root condition memorised with the factors in the wrong pairing — derive by elimination if unsure, or just subtract.
- Two quadratics can share both roots without being identical: proportional coefficients are enough.

STANDARD APPLICATION

The graph view (what the algebra is saying)

$y = ax^2 + bx + c$: vertex $\left(-\frac{b}{2a}, \frac{4ac - b^2}{4a}\right)$, axis $x = -\frac{b}{2a}$, meets the axis iff $D \geq 0$, opens up iff $a > 0$. Shifting roots by k translates the graph; reversing coefficients reflects the root set by $r \mapsto \frac{1}{r}$. Every transformation above is a picture before it is a formula.

LESSON 07 / 08

The five shapes every parameter problem takes

KEY RESULT

Shape recognition

1. **Nature prescribed** ("real/distinct/equal/non-real roots"): write D and solve the inequality — plus the $a \neq 0$ degeneracy check.
2. **Condition between roots** ("one root is double the other", "roots differ by 2", "reciprocals of each other"): Vieta plus the extra relation. Reciprocals condition: $P = 1$, i.e. $c = a$. One root double: roots $t, 2t$, so $S = 3t$, $P = 2t^2$, eliminate: $9P = 2S^2$ (for the monic-scaled S, P).
3. **Sign prescribed** ("both positive", "opposite signs"): sheet 04, Vieta form: $D \geq 0$, signs of S and P.
4. **Location prescribed:** sheet 04 toolkit, three ingredients.
5. **Range/extremum asked:** vertex (sheet 03), or write $y = \frac{\text{quadratic}}{\text{quadratic}}$, cross-multiply to a quadratic in x and demand $D \geq 0$ — the **discriminant method** for ranges.

WORKED EXAMPLE

Worked — discriminant method for a range

Range of $y = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$ ($x \in \mathbb{R}$; denominator > 0 always since its $D = 9 - 16 < 0$). Cross-multiply: $(y-1)x^2 + 3(y+1)x + 4(y-1) = 0$. Real x exists iff $D = 9(y+1)^2 - 16(y-1)^2 \geq 0$, i.e. $-7y^2 + 50y - 7 \geq 0$, i.e. $(7y-1)(y-7) \leq 0$, so $\frac{1}{7} \leq y \leq 7$. (The case $y = 1$ separately gives $x = 0$, consistent.) Equality endpoints are attained, so the range is the closed interval.

KEY RESULT

Recap — the chapter on one card

Formula $x = \frac{-b \pm \sqrt{D}}{2a}$; D cases; $S = -\frac{b}{a}$, $P = \frac{c}{a}$; $\alpha^2 + \beta^2 = S^2 - 2P$; $\alpha^3 + \beta^3 = S^3 - 3PS$; $(\alpha - \beta)^2 = S^2 - 4P = \frac{D}{a^2}$; vertex $\left(-\frac{b}{2a}, \frac{4ac - b^2}{4a}\right)$; sign: a's sign outside, opposite between; always ≥ 0 iff $a > 0, D \leq 0$; location = D + $af(k)$ + vertex; reducible = substitute, filter by range, verify; common root = subtract; transformed = new S, new P.

COMMON TRAP

The ten mistakes that cost the marks (chapter checklist)

1. $D = b^2 + 4ac$ sign slip. 2. "$D < 0 \Rightarrow$ no roots." 3. $(\alpha - \beta)^2 = S^2 - 2P$. 4. Vertex value called max when $a > 0$. 5. Sign pattern ignoring sign of a. 6. Location: dropping one of D, $af(k)$, vertex. 7. Parameter in a: degenerate linear case unchecked. 8. Squared-roots product written P not P^2. 9. No verification after squaring. 10. Dividing away a vanishing factor and losing a root.

INTUITION

Forward map — where quadratics reappear

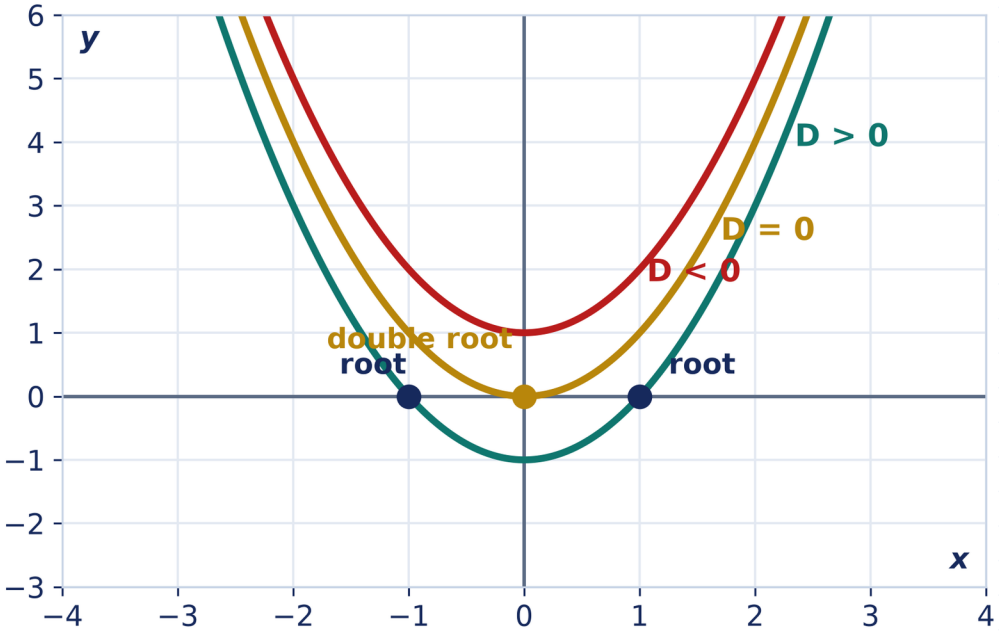
Vieta is the seed of symmetric polynomials (Part II here, and Newton's identities); the vertex is the seed of optimisation and of the discriminant range method used again in conic ranges; roots-of-unity filters, quadratic residues and Vieta jumping all begin with "read this symmetric condition as a quadratic in one variable".

LESSON 08 / 08

The discriminant describes intersections

For $f(x) = ax^2 + bx + c$ with $a \neq 0$, the roots are the x-intercepts. Completing the square gives $f(x) = a((x + b/(2a))^2 - D/(4a^2))$, where $D = b^2 - 4ac$. The sign of D therefore controls two, one, or no real intersections.

Discriminant as an intersection count



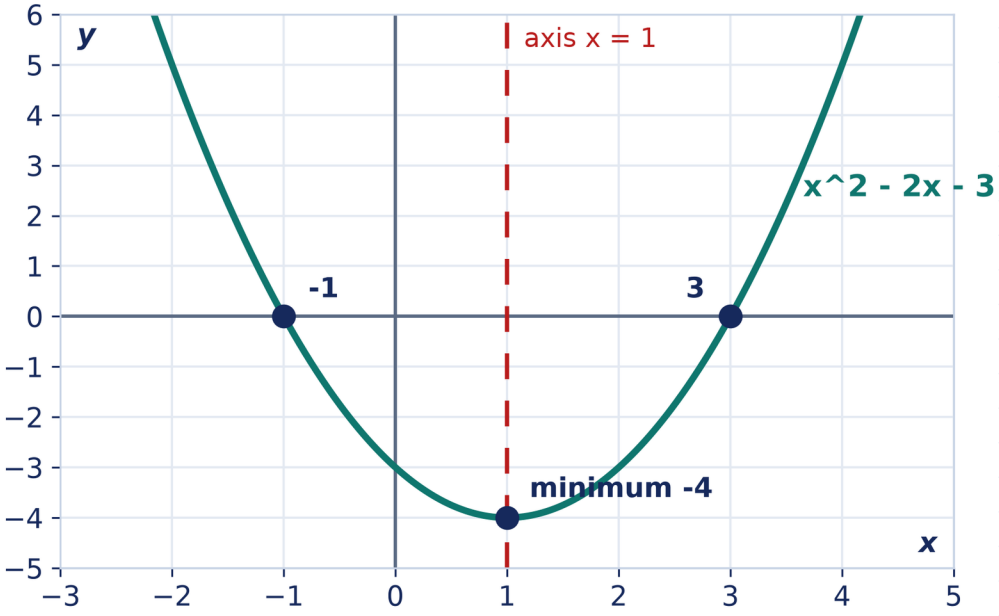
READ THE FIGURE
Vertical translation does not change the leading coefficient or axis. It changes the discriminant and hence the number of real x-intercepts.

When $a > 0$, the parabola opens upward; when $a < 0$, downward. If $D < 0$, there are no real roots and the expression keeps the sign of a for every real x. This is often faster than attempting a sign table with no roots.

The vertex gives the extreme value

Completing the square also gives the vertex $(-b/(2a), -D/(4a))$. For $f(x) = x^2 - 2x - 3 = (x-1)^2 - 4$, the vertex is $(1, -4)$, the axis is $x = 1$, and the range is $[-4, \infty)$.

Vertex, axis, intercepts and range in one picture



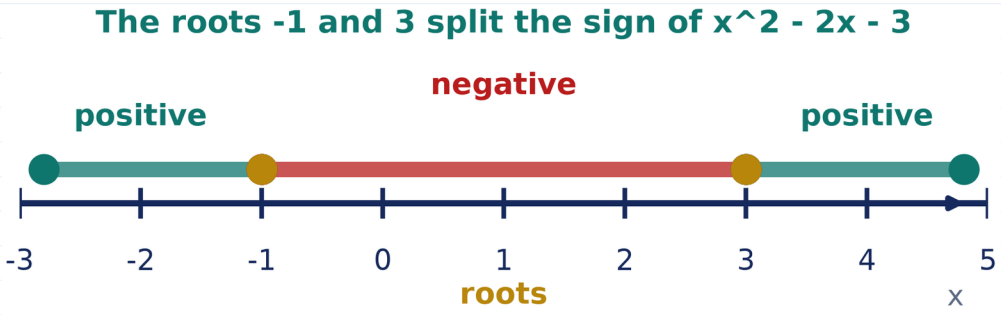
READ THE FIGURE

The roots are symmetric about the axis: their mean is 1. The minimum is attained, so the range includes -4 and every larger real number.

Vieta's relations are visible here: the sum of the roots is twice the axis coordinate, $-1 + 3 = 2 = -b/a$, and their product is $(-1)3 = -3 = c/a$. Use these as structural checks even when exact roots are not required.

Root location is a sign-and-axis problem

A root-location question asks more than whether roots exist. To place both roots of an upward parabola to the right of k, require a nonnegative discriminant, an axis to the right of k, and $f(k) > 0$. For a double root, use the corresponding boundary interpretation carefully.



READ THE FIGURE

An upward quadratic is positive outside its distinct roots and negative between them. If roots are repeated or absent, this alternating pattern must be modified.

For $x^2 - 2x - 3 = (x + 1)(x - 3)$, the signs follow directly from the factors. The graph and factorisation agree: at $x = 0$, $f(0) = -3$, placing the origin in the negative interval.

A worked parameter filter

Consider $x^2 - 2mx + m + 2$ and ask for both roots greater than 1. The discriminant is $4(m-2)(m+1)$, the axis is $x = m$, and $f(1) = 3 - m$. The three conditions become

$$(m-2)(m+1) \geq 0, \qquad m > 1, \qquad 3 - m > 0.$$

Their intersection is $2 \leq m < 3$. At $m = 2$, the double root is $x = 2$, which is allowed; as $m \to 3^-$, one root approaches 1 from above.

Root-location workflow



READ THE FIGURE

At m = 2 the equation is (x - 2)^2. At m = 5/2 the roots are (5 +/- sqrt(7))/2, both greater than 1. Boundary and interior checks confirm the parameter interval.

KEY RESULT

Keep all three conditions

For $m = 5/2$, the polynomial is $x^2 - 5x + 9/2$, whose roots are $(5 \pm \sqrt{7})/2$, both greater than 1. Discriminant alone would allow unwanted parameters; the axis and endpoint-sign conditions are what locate the roots.

LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the official syllabus sentence (quadratics with real coefficients), power-sum recurrences and closed forms for $\alpha^n + \beta^n$ are extension theory. Nothing here is needed for Part I; the habit — never solve for roots when a symmetric combination is asked — is the transferable skill.

Power sums from coefficients alone

SHORT DERIVATION

Newton's recurrence for two roots

α, β satisfy $ax^2 + bx + c = 0$. Multiply the equation for α by α^{n-2}, the one for β by β^{n-2}, and add. With $S_n = \alpha^n + \beta^n$:

$$aS_n + bS_{n-1} + cS_{n-2} = 0, \qquad S_0 = 2, \quad S_1 = -\frac{b}{a}$$

So $S_n = \frac{-bS_{n-1} - cS_{n-2}}{a}$ generates every power sum from the coefficients — no radicals, ever. Start at $S_0 = 2$, not S_1 (checklist item).

WORKED EXAMPLE

Worked — Newton's sums

$x^2 - 5x + 3 = 0$: $S_0 = 2$, $S_1 = 5$; $S_2 = 5 \cdot 5 - 3 \cdot 2 = 19$; $S_3 = 5 \cdot 19 - 3 \cdot 5 = 80$. Matches $S^2 - 2P = 19$ and $S^3 - 3PS = 125 - 45 = 80$ ✓ (and numerically $\alpha \approx 4.3028, \beta \approx 0.6972$ give $\alpha^3 + \beta^3 \approx 80.00$ ✓).

KEY RESULT

The golden quadratic generates Fibonacci and Lucas

For $x^2 - x - 1 = 0$ with roots $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$: induction on $\alpha^2 = \alpha + 1$ gives

$$\alpha^n = F_n\alpha + F_{n-1} \qquad (F_0 = 0, F_1 = 1)$$

for both roots. Adding the two identities:
$S_n = \varphi^n + \psi^n = F_n + 2F_{n-1} = F_{n+1} + F_{n-1} = L_n$, the **Lucas numbers**
$2, 1, 3, 4, 7, 11, 18, 29, 47, \ldots$ Subtracting and dividing by $\sqrt{5} = \varphi - \psi$ gives **Binet**:
$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$. Worked check: $F_{10} = 55$, and $55\varphi + 34 \approx 122.9918 \approx \varphi^{10}$ ✓.

STANDARD APPLICATION

Standard applications

Integer-valuedness: if S, P are integers, every S_n is an integer (recurrence), so $\varphi^n + \psi^n \in \mathbb{Z}$ despite the radicals — the standard trick for proving $\lfloor \varphi^{2n} \rfloor$-type results, using $|\psi| < 1$ so $\psi^n \to 0$ and the integer L_n sits within 1 of φ^n.

COMMON TRAP

Common traps — power sums

- Recurrence sign: $aS_n + bS_{n-1} + cS_{n-2} = 0$ — move terms carefully.
- $S_0 = 2$ (two roots), including when a root is repeated.
- Binet needs $\varphi - \psi = \sqrt{5}$ in the denominator, not $2\sqrt{5}$.

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

The discriminant as $a^2(\alpha-\beta)^2$, rational-root criteria, and the resolvent solution of the cubic are classical extension material beyond the official quadratic-equations syllabus line.

KEY RESULT

What the discriminant actually measures

$$\Delta = b^2 - 4ac = a^2(\alpha-\beta)^2$$

The factor $a^2 > 0$ is pure scale: the **sign** of Δ is the sign of $(\alpha-\beta)^2$ — positive (distinct real), zero (coincident), negative (non-real conjugates, where $\alpha-\beta$ is purely imaginary). The **value** measures root separation. One number, complete information.

KEY RESULT

Rational roots are a perfect-square question

For rational a, b, c: roots rational $\Leftrightarrow$ Δ is the square of a rational. Worked: $x^2 - kx + (k-1) = 0$ has $\Delta = k^2 - 4k + 4 = (k-2)^2$, a perfect square for **every** integer k; roots $x = \frac{k \pm |k-2|}{2} = 1, k-1$ (indeed $x^2 - kx + k - 1 = (x-1)(x-k+1)$ ✓).

SHORT DERIVATION

Lagrange's resolvent — a cubic solved by one quadratic

For $x^3 + px^2 + qx + r = 0$, the shift $x = y - \frac{p}{3}$ kills the square term: depressed cubic $y^3 + Py + Q = 0$. Set $y = u + v$; then $y^3 = u^3 + v^3 + 3uvy$, so matching needs $3uv = -P$ and $u^3 + v^3 = -Q$. Hence $u^3v^3 = -\frac{P^3}{27}$, and u^3, v^3 are the roots of the **resolvent quadratic**

$$t^2 + Qt - \frac{P^3}{27} = 0$$

Solve it, take cube roots chosen with $uv = -\frac{P}{3}$ (the pairing constraint is essential — arbitrary cube roots give wrong y), and $y = u + v$. Worked: $x^3 - 6x^2 + 11x - 6 = 0$: $x = y + 2$ gives $y^3 - y = 0$; resolvent $t^2 + \frac{1}{27} = 0$; the three admissible (u, v) pairings give $y = 1, 0, -1$, so $x = 1, 2, 3$ ✓ (sum $6 = -p$, product $6 = -r$ ✓).

INTUITION

Why the ladder stops at degree 4

Quadratic → cubic (resolvent quadratic) → quartic (resolvent cubic): each degree is solved by the previous. At degree 5 the ladder provably ends — the general quintic is not solvable by radicals (Abel–Ruffini), and the explanation of why is Galois theory. The resolvent is the last rung reachable by pure algebra.

COMMON TRAP

Common traps — resolvents and discriminants

- Taking cube roots of u^3, v^3 independently: the three choices must satisfy $uv = -\frac{p}{3}$.
- "Rational coefficients" hypothesis dropped: Δ a perfect square decides rationality only over $\mathbb{Q}$.
- $\Delta = a^2(\alpha - \beta)^2$, not $(\alpha - \beta)^2$ — the a^2 matters for exact values.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IMO-level)

Vieta jumping is a number-theoretic descent technique, far beyond the JEE syllabus. It is included because its engine is exactly Part I material: Vieta's relations plus root-flipping.

KEY RESULT

The move, isolated

Given a symmetric condition on positive integers (a, b) with a fixed integer quotient k: read the condition as a **quadratic in one variable**. One root is a; Vieta hands you the other root $a' = (\text{sum of roots}) - a = \frac{\text{product}}{a}$, automatically an integer with the same k. If $0 \le a' < b \le a$, the pair (b, a') is a smaller solution with the same k — descend until forced to stop; the terminal configuration reveals what k must be.

SHORT DERIVATION

IMO 1988 Problem 6 — the famous application

Claim: if a, b are positive integers with $ab + 1 \mid a^2 + b^2$, then $k = \frac{a^2+b^2}{ab+1}$ is a perfect square. Fix such a k and view $a^2 - kba + (b^2 - k) = 0$ as a quadratic in a. The other root is $a' = kb - a = \frac{b^2-k}{a}$: an integer since $a \mid b^2 - k$ (rearrange the equation: $b^2 - k = a(kb - a)$). Ordering $A \ge B$, one checks $0 \le a' < B$, so descent strictly shrinks the pair while k never changes. It can only halt at $a' = 0$, where the equation gives $k = B^2$ — a perfect square, and the same k as the original pair. ∎

WORKED EXAMPLE

Worked — tracing the descent

$(a, b) = (8, 30)$: $k = \frac{64+900}{241} = 4$. Order $A = 30 \ge B = 8$: other root $a' = 4 \cdot 8 - 30 = 2$, pair $(8, 2)$ (check $\frac{64+4}{17} = 4$ ✓). Reorder $A = 8, B = 2$: $a' = 4 \cdot 2 - 8 = 0$; halt with $k = 2^2 = 4$ ✓. Brute force over $1 \le a, b < 300$: every admissible quotient is a perfect square $(1, 4, 9, 16, 25, 36, \ldots)$ — no exceptions.

STANDARD APPLICATION

The k = 9 chain (building solutions up)

Descent runs from any solution down to $(0, 3)$ for $k = 9$; reversing $((a, b) \mapsto (b, kb - a))$ builds all solutions upward: $(0, 3) \to (3, 27) \to (27, 240) \to (240, 2133) \to \cdots$ Checks: $\frac{9+729}{82} = 9$ ✓; $\frac{729+57600}{6481} = 9$ ✓. So with $a, b \le 500$: $(3, 27), (27, 3), (27, 240), (240, 27)$.

COMMON TRAP

Common traps — Vieta jumping

- Forgetting k is *invariant* under the flip — that invariance is the entire proof.
- Assuming a' is a non-negative integer without proof: integrality comes from Vieta (product/sum of roots), non-negativity from $a' = \frac{B^2 - k}{A}$ plus the ordering argument; both need care in a written solution.
- Descending without a well-ordering: the descent terminates because positive integers cannot decrease forever — state it.